Efficient Estimation of Airbnb Prices: A Comparison of Simple Random and Stratified Sampling Designs

Taksh Girdhar

1 Introduction

The rapid growth of Airbnb has significantly transformed the short-term rental market by allowing individual hosts to set their own prices in a highly decentralized marketplace (Kennedy et al., 2025). Unlike traditional hotel pricing, which is often guided by established revenue management practices, Airbnb pricing is heterogeneous and influenced by a wide range of listing characteristics such as location, room type, and host behavior (Gibbs et al., 2018). As a result, nightly prices exhibit substantial variability, making it difficult to accurately estimate typical market values.

Understanding the distribution of Airbnb prices is important for multiple stakeholders. For travellers, reliable estimates of average prices help inform budgeting decisions. For policymakers and housing advocates, price levels provide insight into the broader economic impact of short-term rentals, particularly in cities like Vancouver where housing affordability is a major concern (Hidiroglou and Kozak, 2018). However, direct estimation is complicated by large population sizes, high variability, and missing data within publicly available datasets.

This project focuses on estimating the mean nightly Airbnb price in Vancouver using statistical sampling methods. Specifically, the research question is *what is the mean nightly Airbnb price in Vancouver, and how does the choice of sampling design affect the precision of this estimate?*

To address this question, we compare simple random sampling (SRS) and stratified sampling based on neighborhoods, which are known to exhibit meaningful differences in price distributions (Hillson et al., 2015). Given the strong presence of significant variability across listings, stratification is expected to reduce variance and improve estimation accuracy. In addition, simulation methods are used to evaluate estimator performance, including variance, confidence interval coverage, and design efficiency.

Overall, this study demonstrates how classical sampling techniques can be applied to real-world, high-variability data to produce reliable population estimates, while highlighting the practical importance of sampling design in statistical inference.

2 Methods

2.1 Data and target parameter

The analysis treated the cleaned Vancouver Airbnb listings data as a finite population and aimed to estimate the population mean nightly listing price (Inside Airbnb, 2025). After exploratory data analysis, the dataset contained $N = 4{,}702$ listings with observed prices. The study variable was `price`, and `neighborhood` was used as the stratification variable. The final process kept 23 neighborhood strata and used a primary sample size of $n = 300$. The main analysis used the full cleaned price variable, while sensitivity analyses also considered a price cap of $5,000 and the exclusion of observations flagged by an IQR-based outlier indicator.

2.2 Population structure and stratum setup

Before drawing samples, the population was divided into disjoint neighborhood strata. For each stratum h, the code computed the stratum size N_h, stratum mean, stratum standard deviation S_h, and weight $W_h = \frac{N_h}{N}$. Internal checks confirmed that the stratum counts summed to the full population size and that the final allocations summed to the target sample size.

2.3 Simple random sampling & Stratified random sampling

Simple random sampling without replacement was used as a baseline design. For a sample of size n, the estimator of the population mean was the sample mean, $\hat{\bar{Y}} = \bar{y}$. Its variance was estimated using the finite population correction, $\widehat{\mathrm{Var}}(\bar{y}) = (1 - \frac{n}{N})\frac{s^2}{n}$. A 95% confidence interval was constructed using a t-critical value with $n - 1$ degrees of freedom. Stratified sampling was implemented using `neighborhood` as the stratification variable. The combined estimator of the population mean was $\hat{\bar{Y}}_{st} = \sum_h W_h \bar{y}_h$, where $W_h = N_h/N$. The variance estimator was $\widehat{\mathrm{Var}}(\hat{\bar{Y}}_{st}) = \sum_h W_h^2(1-\frac{n_h}{N_h})\frac{s_h^2}{n_h}$. Confidence intervals were constructed using a t-based approach with an effective degrees-of-freedom adjustment.

2.4 Allocation rules

Two allocation strategies were used. Under proportional allocation, $n_h \propto N_h$, while under Neyman allocation, $n_h \propto N_h S_h$. Neyman allocation assigns more observations to strata that are larger and more variable. Integer rounding was applied while preserving the total sample size and ensuring a minimum number of observations per stratum.

2.5 Single-sample comparison

A single realized sample was drawn under each design (SRS, proportional stratification, and Neyman stratification) with total sample size $n = 300$. For each sample, the analysis reported the estimate, standard error, and 95% confidence interval. The design effect (DEFF) compares the variance of an estimator under a given sampling design to the variance under simple random sampling (SRS), and is used as a simple measure of efficiency across designs (Zipf and Valliant, 2026). A single-sample design effect was computed as $\widehat{\mathrm{DEFF}} = \frac{\hat{V}_{\mathrm{design}}}{\hat{V}_{\mathrm{SRS}}}$, to illustrate how each design compares to SRS, although this was not used as the main measure of performance.

2.6 Monte Carlo simulation

Monte Carlo simulation with $B = 1000$ replications was usedto check how the estimators performed over many repeated samples. The procedure was to repeatedly draw samples from the same finite population, compute the estimated mean each time, and summarize the resulting estimates. This allowed the sampling designs to be compared in terms of coverage and efficiency (Mooney, 1997).

In each replication, samples of size $n = 300$ were drawn under SRS, proportional stratification, and Neyman stratification. For each design, the simulation recorded the estimated mean and whether the corresponding 95% confidence interval contained the true population mean. The results were summarized using the distribution of estimates, empirical coverage probability, empirical design effect relative to SRS, and effective sample size.

2.7 Design effect and effective sample size

The effective sample size represents the size of a simple random sample that would produce the same level of precision as the given sampling design, based on the design effect (Zipf and Valliant, 2026). The main efficiency comparison was based on the empirical design effect similar to the single sample design effect in section 2.5, $\mathrm{DEFF}_{emp} = \mathrm{Var}_{sim}(\hat{\bar{Y}}_{\mathrm{design}})/\mathrm{Var}_{sim}(\hat{\bar{Y}}_{\mathrm{SRS}})$. Values below 1 indicate improved precision relative to SRS. The effective sample size was defined as $n_{\mathrm{eff}} = n/\mathrm{DEFF}_{emp}$, which represents the equivalent SRS sample size that would achieve the same variance.

2.8 Bootstrap diagnostic

A percentile bootstrap can be used as a diagnostic to assess confidence interval performance under skewness. Bootstrapping approximates the sampling distribution by repeatedly resampling from the observed data and computing estimates $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$, whose empirical distribution can be used to assess interval performance (Puth et al., 2015). A percentile bootstrap was used as a diagnostic in this case for the SRS mean to assess interval performance under skewness . This was used only to check whether confidence interval coverage improved under resampling.

2.9 Sensitivity analysis & Auxiliary-variable assessment

To assess robustness to extreme values, the analysis was repeated under three scenarios, the main population, a version with prices capped at \$5,000, and a version excluding IQR-flagged outliers. For each scenario, the same sampling designs and comparisons were applied.

Ratio estimation using `number_of_reviews` was considered but not used. The correlation between `price` and `number_of_reviews` was $r = -0.016$, indicating no meaningful relationship. The final analysis therefore focused on SRS, stratified sampling, simulation-based comparison, and sensitivity analysis.

3 Data Analysis

3.1 Population definition and cleaning results

The final analysis treated the cleaned Airbnb listings data as a finite population. After removing listings with missing prices, the analysis dataset contained $N = 4702$ listings, with `neighborhood` used as the stratification variable and a total of 23 strata. The main sample size used throughout the sampling comparisons was $n = 300$. As Table 1 shows, only 3 listings still had missing prices immediately before the final population was defined, so missingness in the final analysis stage was negligible. This means the remaining uncertainty in the report comes primarily from the sampling design rather than missing outcome values.

Table 1: Population definition summary after cleaning.

Metric	Value	Metric	Value
Final population size	4702	Number of strata	23
Primary sample size	300	Missing prices before final drop	3
Outcome variable	`price`	Stratification variable	`neighborhood`

3.2 Exploratory results on nightly price

The exploratory analysis showed that nightly price is strongly right-skewed. As Table 2 shows, the mean nightly price in the cleaned population was 219.76, while the median was much lower at 158.00. The standard deviation was 650.23, and the observed range was from 14.00 to 40,896.00. The quartiles were $Q_1 = 108.00$ and $Q_3 = 239.00$, giving an interquartile range of 131.00. These results indicate that a relatively small number of very expensive listings have a strong effect on the population mean and on the spread of the data.

This skewness is also clear visually. Figure 1 shows the strong concentration of listings at lower prices and the long right tail, while Figure 2 highlights the extent of the upper-tail outliers. Together, the table and figures show why later sections include both a bootstrap diagnostic and a sensitivity analysis.

Table 2: Summary statistics for nightly price in the cleaned population.

Statistic	Value
Mean	219.76
Median	158.00
Standard deviation	650.23
Q_1	108.00
Q_3	239.00
IQR	131.00
Minimum	14.00
Maximum	40,896.00

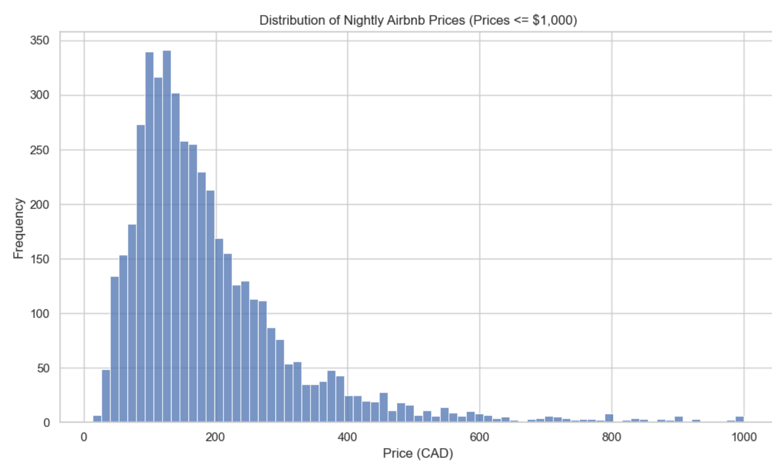

Figure 1: Distribution of nightly Airbnb price in the cleaned population. For display purposes, the histogram is shown only for prices at or below $1,000 so the main shape of the distribution is visible.

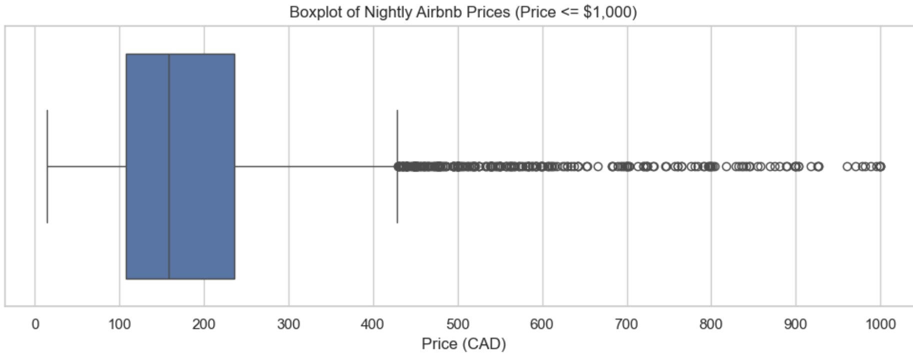

Figure 2: Boxplot of nightly price showing the strong right tail and extreme values. For display purposes, the plot is shown only for prices at or below $1,000 so the central distribution can be seen more clearly.

3.3 Outliers and auxiliary-variable check

Using the IQR rule, the upper outlier fence was 435.50, and 297 listings were flagged as price outliers, which is 6.31% of the final population. As Table 3 shows, this is only a small share of the data, but these listings appear to drive much of the large standard deviation and the instability seen in some interval results.

The EDA also examined whether `number_of_reviews` could be used as a useful auxiliary variable. Figure 3 illustrates that there is no meaningful linear pattern between price and number of reviews, and the estimated correlation was $r = -0.016$. Since this relationship is essentially zero, ratio estimation based on review counts was not pursued further.

Table 3: Outlier summary for nightly price based on the IQR rule.

Quantity	Value
Upper fence	435.50
Listings flagged	297
Percent flagged	6.31%

Figure 3: Price versus number of reviews, showing negligible linear association ($r = -0.016$). For display purposes, the plot is restricted to prices at or below $1,000 so the main pattern is easier to see.

3.4 Neighborhood structure and allocation results

Neighborhood differences provide the main reason for using stratified sampling. As Table 4 shows, the largest stratum was Downtown, with $N_h = 1180$, mean price 300.44, and standard deviation 1241.83. Other large strata were much less variable, such as Kitsilano ($N_h = 321$, mean $= 242.94$, $S_h = 243.39$) and Mount Pleasant ($N_h = 306$, mean $= 165.52$, $S_h = 126.44$). This variation across strata supports the use of neighborhood as a stratification variable. These differences are also reflected in the allocation results. As Table 5 shows, both proportional and Neyman allocation satisfied $\sum_h n_h = 300$. The minimum stratum sample sizes were 3 under proportional allocation and 2 under Neyman allocation. The largest difference occurred in Downtown, where proportional allocation assigned 66 sampled listings while Neyman allocation assigned 175. This shows that Neyman allocation shifted a much larger share of the sample toward the stratum with the highest variability.

Table 4: Example neighborhood-level stratum summaries.

neighborhood	N_h	Mean price	S_h	W_h
Downtown	1180	300.44	1241.83	0.25
Kitsilano	321	242.94	243.39	0.07
Mount Pleasant	306	165.52	126.44	0.07
Downtown Eastside	297	231.64	320.13	0.06
Hastings-Sunrise	268	156.36	127.07	0.06

Table 5: Selected allocation results under proportional and Neyman designs.

neighborhood	Proportional n_h	Neyman n_h
Downtown	66	175
Kitsilano	19	11
Mount Pleasant	19	7
Downtown Eastside	18	13
Hastings-Sunrise	16	6
West End	16	8
Kensington-Cedar Cottage	16	10
Riley Park	13	7
Renfrew-Collingwood	13	4
Dunbar Southlands	11	5

3.5 Single-sample comparison

A single realized sample was first used to illustrate how the three designs can differ in practice. As Table 6 shows, under SRS the estimated mean nightly price was 220.59, with standard error 14.93 and a 95% confidence interval of $[191.21, 249.97]$. Under proportional stratification, the estimate was 238.51, with standard error 25.75 and interval $[187.41, 289.60]$. Under Neyman stratification, the estimate was 198.03, with standard error 10.36 and interval $[176.81, 219.25]$. The corresponding one-run design effects were about 2.97 for proportional stratification and 0.48 for Neyman stratification, using SRS as the baseline.

These results are useful descriptively, but they should not be treated as the main comparison because any one draw can look unusually favorable or unfavorable by chance. For that reason, the long-run evaluation in the Monte Carlo section is more important.

Table 6: Single-sample comparison of SRS, proportional stratification, and Neyman stratification.

Method	Estimate	SE	95% CI	DEFF
SRS	220.59	14.93	[191.21, 249.97]	1.00
Stratified–Proportional	238.51	25.75	[187.41, 289.60]	2.97
Stratified–Neyman	198.03	10.36	[176.81, 219.25]	0.48

3.6 Monte Carlo results

The main design comparison used $B = 1000$ Monte Carlo replications. As Table 7 shows, the true finite-population mean was 219.7561. Under SRS, the empirical coverage of the nominal 95% interval was 0.774. Under proportional stratification, coverage was 0.743, with empirical design effect 0.8907 and implied effective sample size 336.8217. Under Neyman stratification, coverage increased to 0.845, with empirical design effect 0.4681 and implied effective sample size 640.8763.

The simulation results are also clear in the figures. Figure 4 shows the sampling distributions of the estimated mean under the three designs, and Figure 5 illustrates the spread of estimates across replications. Figure 6 shows that all three methods under-covered the true mean relative to the nominal 95% level, although Neyman stratification performed best.

Table 7: Monte Carlo summary of sampling performance by design.

Method	Coverage	DEFF_{emp}	n_{eff}	Relative note
SRS	0.774	1.0000	300.0000	Baseline
Stratified–Proportional	0.743	0.8907	336.8217	Small gain over SRS
Stratified–Neyman	0.845	0.4681	640.8763	Strongest gain

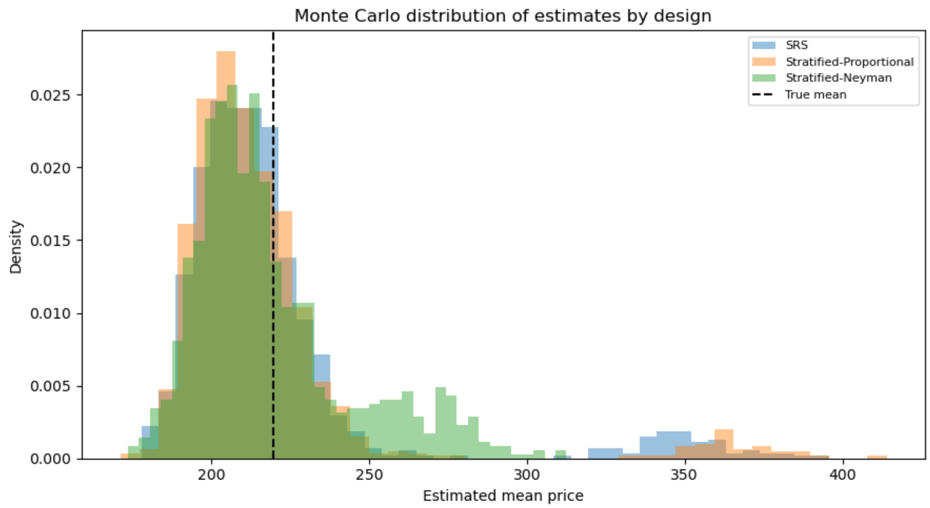

Figure 4: Sampling distributions of the estimated mean under SRS, proportional stratification, and Neyman stratification.

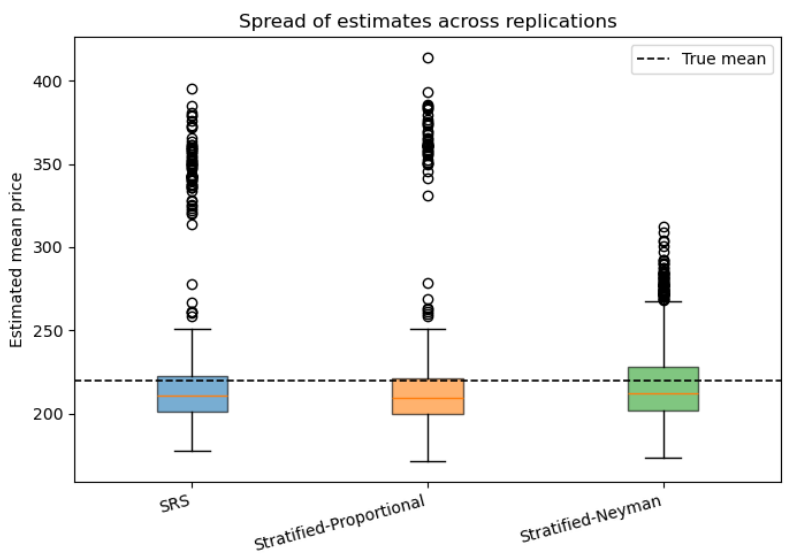

Figure 5: Spread of estimated means across Monte Carlo replications for each sampling design.

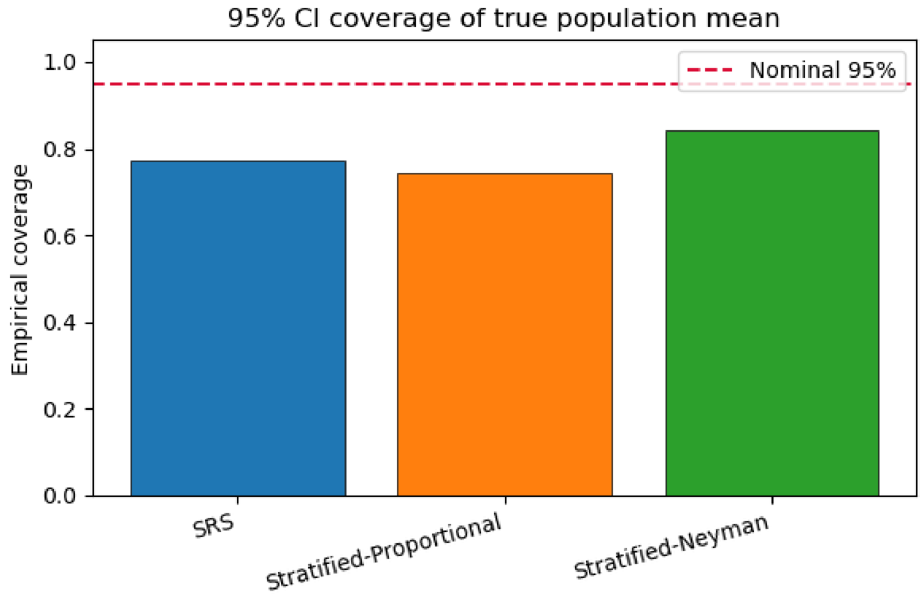

Figure 6: Empirical confidence interval coverage by design, with the nominal 95% level shown for reference.

Overall, these results show that Neyman allocation was the most efficient design in this project. Relative to SRS, its variance was about 53% lower, and its implied effective sample size was much larger than the nominal $n = 300$. Proportional allocation also improved on SRS, but only modestly. At the same time, all three methods under-covered the true mean. The skewed price distribution makes it harder for confidence intervals to achieve the nominal 95% coverage.

3.7 Bootstrap diagnostic

A percentile bootstrap diagnostic was also run for the SRS mean. Across 300 simulated samples, with 400 bootstrap resamples per sample, the bootstrap coverage was 0.793, which was still below the nominal 0.95 level.

This supports the same main point as the Monte Carlo results shown in Table 7 and Figure 6: the skewed price distribution makes it harder for confidence intervals to achieve the nominal 95% coverage.

3.8 Sensitivity analysis

To check how much the results depended on extreme prices, the analysis was repeated under three scenarios: the main cleaned population, a version with prices capped at $5,000, and a version excluding IQR-flagged outliers. As Table 8 shows, in the main population the estimates were 225.43 for SRS, 181.47 for proportional stratification, and 220.02 for Neyman stratification. Under the $5,000 cap, the estimates were 201.87, 220.05, and 219.24, respectively. When IQR-flagged outliers were excluded, the estimates dropped further to 170.23, 166.16, and 167.53.

Figure 7 illustrates the same pattern visually. The corresponding confidence intervals in Table 8 also became noticeably tighter after capping or removing extreme prices. These results connect directly back to the EDA findings in Table 2, Figure 1, and Figure 2. A relatively small fraction of listings in the extreme upper tail has a large effect on both the population mean and the sampling variability. The sensitivity analysis therefore helps explain why the uncapped mean is unstable and why interval coverage was weaker than expected in the Monte Carlo study.

Table 8: Sensitivity analysis across the main, capped, and outlier-excluded populations.

Scenario	Method	Estimate with 95% CI	
Main	SRS	225.43	[188.25, 262.60]
Main	Stratified–Proportional	181.47	[167.85, 195.09]
Main	Stratified–Neyman	220.02	[191.12, 248.92]
Cap at $5,000	SRS	201.87	[185.64, 218.10]
Cap at $5,000	Stratified–Proportional	220.05	[189.35, 250.75]
Cap at $5,000	Stratified–Neyman	219.24	[195.73, 242.74]
Exclude IQR outliers	SRS	170.23	[160.32, 180.14]
Exclude IQR outliers	Stratified–Proportional	166.16	[157.30, 175.03]
Exclude IQR outliers	Stratified–Neyman	167.53	[158.62, 176.44]

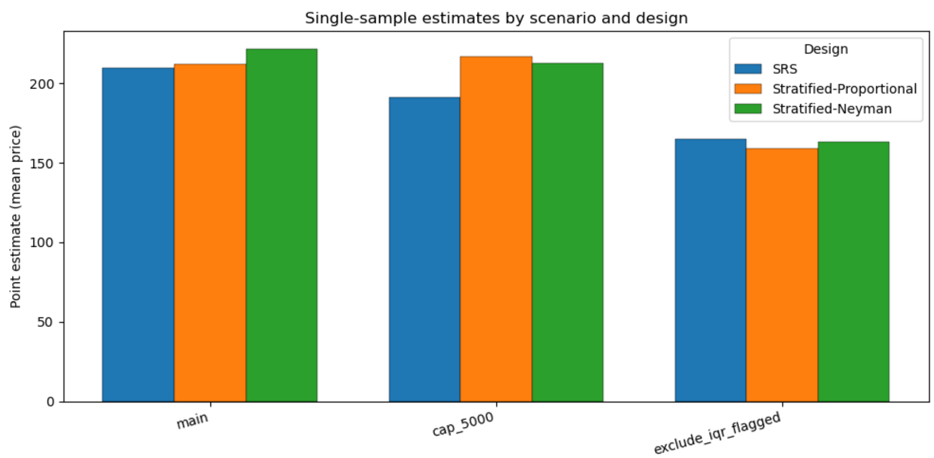

Figure 7: Comparison of point estimates across sensitivity scenarios.

3.9 Summary of findings

Overall, the data analysis points to three main findings. First, as Table 2, Figure 1, and Figure 2 show, nightly Airbnb prices in Vancouver are highly right-skewed, and a small fraction of extreme listings strongly affects the mean. Second, as Table 4, Table 5, and Table 7 show, neighborhood is a useful stratification variable, and Neyman allocation gives the clearest gain in efficiency over both SRS and proportional stratification. Third, as Figure 6 and Table 8 indicate, although stratification improves precision, interval coverage remains below nominal because the heavy upper tail of the price distribution continues to affect uncertainty estimation. The bootstrap and sensitivity results both reinforce that conclusion.

4 Conclusion and Discussion

This project evaluated how different sampling designs affect the estimation of the mean nightly Airbnb price using a finite population of Vancouver listings. The results consistently show that stratified sampling improves estimation efficiency compared to simple random sampling (SRS), with Neyman allocation providing the strongest performance.

The exploratory analysis established that the price distribution is highly right-skewed, with a small proportion of extreme listings exerting a strong influence on both the mean and the variance. As shown in Table 2 and Figures 1–2, the gap between the mean and median, along with the long upper tail, indicates that the population is not well-behaved for standard normal-based inference. This directly explains the undercoverage observed in the simulation results.

The Monte Carlo study confirmed that sampling design plays a major role in efficiency. As Table 7 and Figure 4 show, Neyman allocation substantially reduced variance relative to SRS, with an empirical design effect well below 1. This implies that the same level of precision could be achieved with a much smaller sample size under Neyman allocation. Proportional allocation also improved on SRS, but to a lesser extent. However, Figure 6 shows that all methods failed to achieve nominal 95% coverage, indicating that distributional features of the data, rather than sampling design alone, limit inference accuracy.

The bootstrap diagnostic supported this conclusion by showing that even resampling-based intervals did not fully correct the coverage issue. This reinforces that the skewed distribution of prices is a primary challenge for interval estimation in this context.

The sensitivity analysis further clarified the role of extreme values. As Table 8 and Figure 7 show, both capping extreme prices and removing IQR-based outliers led to noticeably lower estimates of the mean and tighter confidence intervals. This demonstrates that a relatively small fraction of listings in the upper tail drives much of the instability in both point and interval estimates. These findings are consistent with the patterns observed earlier in the EDA.

From a practical perspective, the results suggest that stratified sampling by neighborhood, especially using Neyman allocation, is an effective approach for estimating average Airbnb prices when there is substantial heterogeneity across areas. This approach would be particularly useful for analysts, urban planners, or policymakers who need reliable estimates of short-term rental prices across different parts of a city.

At the same time, the project highlights an important limitation that standard confidence interval methods may perform poorly when the outcome variable is highly skewed. Even with improved sampling designs, the heavy-tailed nature of the data can lead to undercoverage. In practice, this suggests that analysts should consider data transformations, alternative estimators, or robust inference methods when working with similarly skewed economic data.

Overall, this project demonstrates that both sampling design and data characteristics must be considered together. While stratification improves efficiency, the underlying distribution ultimately determines how well statistical inference performs.

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A Appendix: Core Algorithms Used in the Analysis

This appendix summarizes the main code components used in the data analysis. Only the algorithms directly relevant to the reported results are included. Visit the Github repository for the full code.

Algorithm 1: Simple random sampling (SRS) estimator for the population mean

Input: Sample s of size n, population size N, study variable y
Output: $\hat{\bar{Y}}$, $\widehat{\mathrm{Var}}(\hat{\bar{Y}})$, SE, 95% CI

Compute $\bar{y} \leftarrow \frac{1}{n}\sum_{i\in s} y_i$;
Compute $s^2 \leftarrow \frac{1}{n-1}\sum_{i\in s}(y_i - \bar{y})^2$;
Compute sampling fraction $f \leftarrow n/N$;
Compute variance estimate$\widehat{\mathrm{Var}}(\hat{\bar{Y}}) \leftarrow (1-f)\frac{s^2}{n}$;
Compute SE $\leftarrow \sqrt{\widehat{\mathrm{Var}}(\hat{\bar{Y}})}$;
Compute $t^\star$ using $n-1$ degrees of freedom;
Construct 95% CI: $\bar{y} \pm t^\star\mathrm{SE}$

Algorithm 2: Build neighborhood-level stratum summary statistics

Input: Cleaned population data frame, stratum variable `neighborhood`, outcome `price`
Output: Table of N_h, $\bar{Y}_h$, S_h, and W_h for all strata

Group the population by neighborhood;
For each stratum h, compute: $N_h, \quad \bar{Y}_h, \quad S_h$;
Compute total population size N;
For each stratum, compute $W_h \leftarrow \frac{N_h}{N}$;
Return the stratum summary table sorted by N_h;

Algorithm 3: Rounded allocation with minimum sample size per stratum

Input: Allocation weights w_h, total sample size n, minimum per stratum m
Output: Integer allocation n_h with $\sum_h n_h = n$

Set base allocation $n_h \leftarrow m$ for all strata;
Compute remaining sample size: $r \leftarrow n - Hm$;
Distribute the remaining sample proportionally using w_h;
Take floor values of provisional allocations;
Assign leftover units to strata with largest fractional remainders;
Ensure final allocation satisfies $\sum_h n_h = n$;

Algorithm 4: Proportional allocation

Input: Stratum summary table, total sample size n
Output: Allocation n_h

Set weights $w_h \leftarrow N_h$;
Apply Algorithm 3 to obtain integer n_h;

Algorithm 5: Neyman allocation

Input: Stratum summary table, total sample size n
Output: Allocation n_h

Set weights $w_h \leftarrow N_h S_h$;
If any $N_h S_h = 0$, replace zero values with a small positive fallback;
Apply Algorithm 3 to obtain integer n_h;

Algorithm 6: Draw a stratified sample

Input: Population data, allocation vector n_h, stratum variable `neighborhood`
Output: Stratified sample s_{st}

Initialize empty sample;
  For each stratum h: Subset the population to neighborhood h;
    Draw n_h units without replacement;
    Append selected units to the final sample;
Concatenate all selected stratum samples;
Return the combined sample;

Algorithm 7: Stratified estimator of the population mean

Input: Stratified sample s_{st}, stratum summary table, population size N
Output: $\hat{\bar{Y}}_{st}$, $\widehat{\mathrm{Var}}(\hat{\bar{Y}}_{st})$, SE, 95% CI

For each sampled stratum h, compute: $n_h, \quad \bar{y}_h, \quad s_h^2$;
Join sample summaries to the population stratum table;
Compute $W_h = \frac{N_h}{N}, \qquad f_h = \frac{n_h}{N_h}$;
Compute the stratified mean estimator: $\hat{\bar{Y}}_{st} \leftarrow \sum_h W_h \bar{y}_h$;
Compute the variance estimator: $\widehat{\mathrm{Var}}(\hat{\bar{Y}}_{st}) \leftarrow \sum_h W_h^2 (1 - f_h) \frac{s_h^2}{n_h}$;
Compute SE $\leftarrow \sqrt{\widehat{\mathrm{Var}}(\hat{\bar{Y}}_{st})}$;
Approximate effective degrees of freedom using the stratum variance components;
Construct 95% CI: $\hat{\bar{Y}}_{st} \pm t^\star \mathrm{SE}$;

Algorithm 8: Single-sample comparison across designs

Input: Population data, stratum summary table, proportional allocation, Neyman allocation, $n = 300$
Output: One-run estimates, standard errors, CIs, and single-sample DEFF values

Draw one SRS sample of size n;
Draw one proportional stratified sample;
Draw one Neyman stratified sample;
Compute SRS estimate using Algorithm 1;
Compute proportional and Neyman estimates using Algorithm 7;
For each stratified design, compute single-sample design effect:

$$\widehat{\mathrm{DEFF}} \leftarrow \frac{\widehat{V}_{\mathrm{design}}}{\widehat{V}_{\mathrm{SRS}}}$$

Store estimate, standard error, CI, and DEFF for all three designs;

Algorithm 9: Monte Carlo comparison of SRS, proportional, and Neyman designs

Input: Population data, stratum summary table, proportional allocation, Neyman allocation, total sample size n, replications B
Output: Monte Carlo summary table and empirical DEFF values

Compute the true population mean: $\bar{Y} \leftarrow \frac{1}{N} \sum_{i=1}^{N} y_i$;
Initialize storage for replication-level estimates and variance ratios;
For $b = 1, \ldots, B$ Draw one SRS sample of size n;
Draw one proportional stratified sample;
Draw one Neyman stratified sample;
Compute SRS estimate using Algorithm 1;
Compute proportional and Neyman estimates using Algorithm 7;
For each design, record: point estimate;
estimated variance;
indicator for whether the 95% CI contains $\bar{Y}$;
Compute per-replication DEFF ratios for proportional and Neyman:

$$\widehat{\mathrm{DEFF}}^{(b)}_{\mathrm{prop}} = \frac{\widehat{V}^{(b)}_{\mathrm{prop}}}{\widehat{V}^{(b)}_{\mathrm{SRS}}}, \qquad \widehat{\mathrm{DEFF}}^{(b)}_{\mathrm{Neyman}} = \frac{\widehat{V}^{(b)}_{\mathrm{Neyman}}}{\widehat{V}^{(b)}_{\mathrm{SRS}}}$$

Aggregate results by design to obtain: empirical variance, coverage, mean estimate Compute empirical design effect:

$$\mathrm{DEFF}_{emp} = \frac{\mathrm{Var}_{sim}(\hat{\bar{Y}}_{\mathrm{design}})}{\mathrm{Var}_{sim}(\hat{\bar{Y}}_{\mathrm{SRS}})}$$

Compute effective sample size: $n_{eff} = \frac{n}{\mathrm{DEFF}_{emp}}$;
Return replication-level and summary-level outputs;

Algorithm 10: Bootstrap diagnostic for SRS interval performance

Input: Population data, SRS sample size n, number of outer replications B, number of bootstrap resamples R
Output: Bootstrap coverage estimate
Compute the true population mean $\bar{Y}$;
Initialize coverage counter;
For $b = 1, \ldots, B$ Draw one SRS sample of size n from the population;
For $r = 1, \ldots, R$ Draw one bootstrap sample with replacement from the SRS sample;
Compute the bootstrap sample mean;
Use the empirical bootstrap quantiles to form a percentile interval;
Check whether the interval contains $\bar{Y}$;
Store the coverage indicator;
Compute bootstrap coverage as the mean of the stored indicators;

Algorithm 11: Sensitivity analysis across alternative price scenarios

Input: Population data, sample size n, random number generator
Output: Scenario-specific estimates, standard errors, and confidence intervals
Create three analysis populations: Main population;
Capped population with

$$y_i^{cap} = \min(y_i, 5000)$$

Trimmed population excluding IQR-flagged outliers;
For each scenario: Rebuild stratum summary statistics;
Compute proportional and Neyman allocations;
Draw one SRS sample, one proportional stratified sample, and one Neyman stratified sample;
Estimate the mean and CI under each design;
Store scenario, method, estimate, standard error, and CI;
Concatenate all scenario-specific outputs into one summary table;